Active Maintenance Report: coppock_etal_2017

Coppock, Gerber, Green and Kern (2017, Political Analysis): Combining Double Sampling and Bounds to Address Nonignorable Missing Outcomes in Randomized Experiments

2026-08-01

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This repository holds the actively maintained replication code for Coppock, Gerber, Green and Kern (2017), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/pan.2016.6
Replication archive	10.7910/DVN/AQB4MP
Preanalysis plan	osf.io/wkj8v

The data are not redistributed here. The deposit is 231 KB across seven files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the hand extraction of the article, `build_ground_truth.R` generates the comparison and runs the coverage gate, and `coppock_etal_2017_ground_truth.csv` is what it writes. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_etal_2017.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its seven files, installs the `attrition` package from the deposited source tarball if it is not already present, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, modelsummary, knitr, kableExtra, gridExtra, here, and `attrition`, which `install_attrition.R` handles. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about seven seconds. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The three PDF figures differ on every run because a PDF records the time it was written; the CSV, TeX and PNG output comes back byte-identical.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Not without intervention, and the obstacle is a bug in the deposit rather than anything that has aged. `experimental_analysis_main_text.R` prints Tables 1, 2 and 3 and then stops at a line that subtracts rows from an object called `mturk_results`, which the script never creates. The object it means is `table_3`, assembled four lines earlier under a different name. Everything after that line, which is the 65 percent confidence interval reduction the results section quotes and the entire sensitivity analysis behind Figure 2, is unreachable. That leaves 3 recorded claims with no archive value at all, all of them numbers the article prints: the reduction in interval width, the sensitivity parameter as Section 4.3 states it, and the same parameter as Figure 2 prints it on its face.

The other two scripts run. `simulation_analysis.R` produces Figure 1 and Appendix Figure 1 with two deprecation warnings and no errors. `experimental_analysis_appendix.R` runs and reproduces Appendix Table 1 exactly, but only against the `commarobust` package as deposited. The current version of `commarobust` on GitHub also calls itself 0.1.0 and exports nothing at all: it has been reduced to a message telling the user to install `estimatr` instead. A reader who follows the archive's own README, which recommends `devtools::install_github()` over the bundled tarball, gets a package that fails on the first line that uses it, and `packageVersion()` reports the same number either way.

The remaining 128 claims the archive can be compared on all reproduce, including every cell of all five published tables.

1.2 Does the maintained rewrite reproduce the paper?

Yes, without exception. Of the 171 recorded claims, 158 carry a published number and all 158 match it to the precision the page prints; 0 differ. The remaining 13 are claims about shape rather than value, and all 13 hold. The three the archive cannot reach are among the matches: replacing the stale object name gives a 65.0 percent reduction in confidence interval width where the paper says 65 percent, and the sensitivity analysis gives a δ^* of 0.0707 where the paper and Figure 2 both say 0.07.

No sentence in the article is contradicted by the data behind it, so this repository carries no errata document.

2 Paper overview

Citation: Coppock, A., Gerber, A. S., Green, D. P. and Kern, H. L. (2017). "Combining double sampling and bounds to address nonignorable missing outcomes in randomized experiments." *Political Analysis*, 25(2), 188-206. DOI: 10.1017/pan.2016.6

Summary: Missing outcomes break the guarantee randomization provides, and the usual repairs assume that missingness is ignorable. The paper proposes an alternative that does not: combine a double sampling design, in which a random subset of the initial nonrespondents is pursued with a larger incentive, with nonparametric worst-case bounds on the outcomes still missing after that second effort. It derives the bounds estimator under double sampling, gives analytic variances and confidence intervals, adds poststratification for covariate adjustment, and adds a sensitivity analysis indexed by the fraction δ of remaining nonrespondents for whom ignorability fails. Monte Carlo simulations show how the identification region and the confidence interval respond to the two response rates. The application is a two-wave Mechanical Turk replication of Levendusky and Malhotra (2016) in which 1,980 subjects read either a polarized or a moderate news story, 1,444 responded in the initial Wave 2 sample, and 50 nonrespondents per arm were offered four times the original incentive, of whom 72 complied. Worst-case bounds from the initial sample alone are $(-1.54, 1.71)$ and say nothing. Double sampling with poststratification narrows the associated confidence interval from 3.50 to 1.23, a 65 percent reduction, while the sensitivity analysis shows the significant naive estimate collapses once ignorability fails for even 7 percent of the remaining nonrespondents.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
experimental_analysis_main_text.R	Prints Tables 1, 2 and 3, then stops: object 'mturk_results' not found	The object meant is table_3, built four lines above under another name
experimental_analysis_appendix.R	Runs against the deposited commarobust; fails against the current one	install.packages('original/commarobust_0.1.0.tar.gz', repos = NULL, type = 'source')
simulation_analysis.R	Runs; two deprecation warnings	guide = FALSE becomes guide = 'none'; %+% becomes +

The deposit is unusually well prepared in one respect and unusually exposed in another. It bundles the source tarballs of both packages the analysis depends on, `attrition` and `commarobust`, neither of which has ever been on CRAN. That is the right instinct, and it is what makes the appendix script reproducible today. The README then undercuts it: it recommends installing from GitHub instead, calling that route "easy," and offers the tarballs only "in the interest of full replicability." Ten years on, the easy route is the broken one.

3.1 The two packages

`attrition` implements the paper's method and is still on GitHub. The deposited version is 0.0.0.9000 and the current one is 1.0.0, and the two no longer agree everywhere. 1.0.0 fixed the Imbens-Manski critical value, which had been found by minimising an absolute value over a fixed interval that only covers alpha near 0.05, so the interval bounds move by about 2e-09 relative here and would move much more at another significance level. Every point estimate is unchanged. 1.0.0 also renamed the six quantities its estimators return, onto broom's names, and reports the uncertainty as a standard error where 0.0.0.9000 reported a variance. The maintained rewrite is written against the current package and `install_attrition.R` halts if an older one is installed, since the old names would otherwise fill these tables with `NA` rather than fail. The deposited tarball is still what the scripts in `original/` are written against, and it still runs them; `install_attrition.R` installs it when `attrition` is absent, because the tarball's checksum is pinned in `original_manifest.csv` and a GitHub repository's contents are not.

`commarobust` is a different case. It was the author's prototype for robust standard errors around `lm()`, and it was superseded by `estimatr`, whose `lm_robust()` defaults to the same HC2 estimator. The current GitHub `commarobust` has been emptied out to a startup message pointing at `estimatr`, and it exports no functions. It nevertheless still calls itself version 0.1.0, exactly like the deposited tarball, which exports six. Two packages with the same name and the same version number differ in whether the archive runs, and nothing in `sessionInfo()` would tell a reader which one they have.

3.2 The stale object name

The main text script builds Table 3 and then reaches for it under a name it does not have:

```
table_3 <- cbind(no_ds = cis_1, ds = cis_2, ds_ps = cis_3)
# ... four lines of row labels and a print ...
widths <- mturk_results[2,] - mturk_results[1,]
(widths[1] - widths[3]) / widths[1]
```

`mturk_results` is never assigned anywhere in the archive. The name presumably survived an interactive session in which the object had been called that, and the rename was not carried through before the file was deposited. This is the class of defect that only a clean-session run finds: in the author's own session the old object was still in the workspace, so the script worked.

The cost is not cosmetic. The line that fails computes the 65 percent reduction in confidence interval width, which is the sentence the results section builds toward, and everything below it, including the entire sensitivity analysis and Figure 2, never executes. Both values are correct once the name is fixed, so the archive contains a correct analysis that it cannot run.

4 Errata

The maintained rewrite makes one correction to the archive's code: `mturk_results` becomes the object that was meant. Nothing else in the estimator, the specification or the sample is changed.

Table 2: The published claims the deposited script cannot reach, and what the rewrite produces.

Claim	Published	Archive	Rewrite
Section 4.2 reduction in 95% CI width	65 percent	not computed, script stops	0.6503
Section 4.3 sensitivity parameter	0.07	not computed, script stops	0.0707
Figure 2 annotation	0.07	not computed, script stops	0.0707

No published value changes. The correction recovers numbers the archive was unable to produce; all of them agree with the article. That is a correction to the deposited code, which is what this section is for. It is not an erratum in the article, and there are none: every number the article prints is reproduced here, and every claim it makes about shape holds.

5 The extraction and the two instruments

Two files stand between the article and this pipeline, and they are deliberately separate.

`ground_truth/published_claims.csv` is the **extraction**: every numeric token in the article and the online appendix, read out by hand and classified. It carries 212 rows. Citation years, equation numbers, footnote markers, section cross-references and page numbers are excluded as typography rather than claims; everything else is here, including numbers spelled out in words ("three scenarios", "two decades", "the four policy questions") that no token scan finds. `value_paper` is the string the page prints, with its digits intact, and `digits` records that precision for each claim separately, because the appendix regression table prints one R-squared at three decimals and its neighbour at four.

Table 3: The extraction, by claim type.

Claim type	Needs a claim block	Rows
definitional	no	26
definitional	yes	16
descriptive	yes	13
pipeline	yes	138
structural	yes	4
transcribed	no	15

A block is required for 171 of those rows. The 41 that do not need one are quantities the deposit was never going to record and that no code can therefore check: the $1.00 and $4.00 payments, the ten days between waves, the seven-point item scales, and the percentages inside the treatment articles themselves, which are quoted from the source study. They are verified against the article's own methods section and at the point of use in the scripts, and they are named here so that a reader can see the boundary rather than infer it. The simulation's outcome means and variances are in the same class: they are inputs to `attrition:::ds_manski_cis_2s()` that no output file carries, and they sit beside the call in `figure_1_simulation.R`.

`maintained/in_text_claims.R` is the **second instrument**. It reads only `maintained/output/`, never the ground truth, and it recomputes each claimed number by its own path: the prose figures of Section 4.2 are derived there from `table_3_bounds_main.csv` and `table_a1_mipo_effects.csv` rather than read out of `text_section_4_claims.csv`, which is the file the ground truth reads. It prints one line per claim, and `ground_truth/build_ground_truth.R` runs it as a program, in its own environment, and asserts three things: that the number of claims printed equals the number the extraction requires, that the two sets of identifiers are equal in both directions, and that every printed value equals the ground truth's own value rendered at the same precision.

Two of the article's claims are checkable only because a second quantity can be recovered from the tables the article prints. Section 4.2 says missing outcomes are replaced with the lowest possible value of 0 and the highest of 6. Neither limit appears in any output file, but with the response rates of Table 1 and the initial-responder means of Table 2, the worst-case low and high estimates of Table 3 are two linear equations in exactly those two limits, and solving them returns 0 and 6.

Coverage of the published floats is complete for the tables: 15 of 15 cells in Table 1, 12 of 12 in Table 2, 18 of 18 in Table 3, 32 of 32 in Appendix Table 1 and 36 of 36 in Appendix Table 2, or 113 published cells with 113 rows. The three figures print no estimates except the $\delta^* = 0.07$ that Figure 2 carries on its face, which is covered. What the figures do assert is shape, and those assertions are covered by claims with computed truth values rather than by cells: that the identification region collapses to a point when everyone responds to the follow-up, that at a follow-up response rate of zero its width returns to the worst-case width of $2(1 - p_1)$, that it narrows as the initial response rate rises, and that raising the sample size a hundredfold shrinks the confidence interval while leaving the identification region unchanged to the last decimal.

6 Number-by-number comparison

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity, or that no archive counterpart to that quantity was measured.

Location	Claim	Paper	Archive	Match
appendix_section_1_p2	appx_n_large	80000		
appendix_section_1_p2	appx_n_small	800		
appendix_section_1_p2	appx_n2_large	5000		
appendix_section_1_p2	appx_n2_small	50		
appendix_section_1_p2	appx_size_factor	100		
appendix_section_1_p2	appx_two_scenarios	2		
appendix_section_2_p3	appx_effect_diminished			
appendix_section_2_p3	appx_extremity_negligible			
appendix_section_2_p3	appx_wave1_effect	0.468		
appendix_section_2_p3	appx_wave1_effect_strong			
appendix_section_2_p3	appx_wave2_effect	0.126		
appendix_section_3_p4	appx_pairwise_columns	6		
figure_1	sim_p2_zero_is_wcb			
figure_1	sim_point_identified			
figure_1	sim_region_smaller_in_p1			
figure_1	sim_width_shrinks_to_zero			
figure_2	fig2_delta_star	0.07		0
figure_a1	appx_ci_shrinks_region_does_not			
section_3_p11	sim_n_followup	50		
section_3_p11	sim_n_total	800		
section_3_p11	sim_p1_max	0.75		
section_3_p11	sim_p1_min	0.25		
section_3_p11	sim_p2_max	1		
section_3_p11	sim_p2_min	0		
section_3_p11	sim_true_ate	0		
section_3_p12	sim_n_scenarios	3		
section_3_p12	sim_p1_scenario_high	0.75		
section_3_p12	sim_p1_scenario_low	0.25		
section_3_p12	sim_p1_scenario_mid	0.50		
section_4.1_p14	app_n_completed	72		
section_4.1_p14	app_n_invited	100		
section_4.1_p14	app_n_invited_per_arm	50		
section_4.1_p14	app_n_responded	1444		
section_4.1_p14	app_n_subjects	1980		
section_4.2_p14	app_outcome_max	6		
section_4.2_p14	app_outcome_min	0		
section_4.2_p14	txt_ci_width_ds_ps	1.23	1.23	1
section_4.2_p14	txt_ci_width_no_ds	3.50	3.50	1
section_4.2_p14	txt_ds_ps_ci_lower	-0.53	-0.53	1
section_4.2_p14	txt_ds_ps_ci_upper	0.70	0.70	1
section_4.2_p14	txt_ds_ps_wcb_high	0.53	0.53	1
section_4.2_p14	txt_ds_ps_wcb_low	-0.34	-0.34	1
section_4.2_p14	txt_ds_wcb_high	0.57	0.57	1
section_4.2_p14	txt_ds_wcb_low	-0.34	-0.34	1
section_4.2_p14	txt_naive_ate_coef	0.126	0.126	1
section_4.2_p14	txt_naive_ate_pval	0.057	0.057	1
section_4.2_p14	txt_naive_ate_se	0.066	0.066	1
section_4.2_p14	txt_no_ds_ci_lower	-1.67	-1.67	1
section_4.2_p14	txt_no_ds_ci_upper	1.84	1.84	1
section_4.2_p14	txt_no_ds_wcb_high	1.71	1.71	1
section_4.2_p14	txt_no_ds_wcb_low	-1.54	-1.54	1
section_4.2_p14	txt_reduction_pct	0.65		0
section_4.3_p15	app_naive_sig_10			
section_4.3_p15	app_no_delta_star_05			
section_4.3_p15	txt_delta_star	0.07		0
table_1	t1_moderate_did_not_respond	264	264	1
table_1	t1_moderate_invited	50	50	1
table_1	t1_moderate_n_initial	995	995	1
table_1	t1_moderate_responded	731	731	1
table_1	t1_moderate_responded_to_invitation	39	39	1
table_1	t1_polarized_did_not_respond	272	272	1
table_1	t1_polarized_invited	50	50	1
table_1	t1_polarized_n_initial	985	985	1
table_1	t1_polarized_responded	713	713	1

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity, or that no archive counterpart to that quantity was measured. *(continued)*

Location	Claim	Paper	Archive	Match
table_1	t1_polarized_responded_to_invitation	33	33	1
table_1	t1_total_did_not_respond	536	536	1
table_1	t1_total_invited	100	100	1
table_1	t1_total_n_initial	1980	1980	1
table_1	t1_total_responded	1444	1444	1
table_1	t1_total_responded_to_invitation	72	72	1
table_2	app_samples_similar			
table_2	t2_moderate_ds_mean	3.583	3.583	1
table_2	t2_moderate_ds_n	39	39	1
table_2	t2_moderate_ds_sd	1.367	1.367	1
table_2	t2_moderate_ir_mean	3.542	3.542	1
table_2	t2_moderate_ir_n	731	731	1
table_2	t2_moderate_ir_sd	1.243	1.243	1
table_2	t2_polarized_ds_mean	3.826	3.826	1
table_2	t2_polarized_ds_n	33	33	1
table_2	t2_polarized_ds_sd	1.313	1.313	1
table_2	t2_polarized_ir_mean	3.668	3.668	1
table_2	t2_polarized_ir_n	713	713	1
table_2	t2_polarized_ir_sd	1.270	1.270	1
table_3	app_ds_narrows			
table_3	app_wcb_uninformative			
table_3	t3_ds_ci_lower	-0.5283	-0.5283	1
table_3	t3_ds_ci_upper	0.7452	0.7452	1
table_3	t3_ds_ps_ci_lower	-0.5290	-0.5290	1
table_3	t3_ds_ps_ci_upper	0.6966	0.6966	1
table_3	t3_ds_ps_var_high	0.0108	0.0108	1
table_3	t3_ds_ps_var_low	0.0126	0.0126	1
table_3	t3_ds_ps_wcb_high	0.5257	0.5257	1
table_3	t3_ds_ps_wcb_low	-0.3444	-0.3444	1
table_3	t3_ds_var_high	0.0111	0.0111	1
table_3	t3_ds_var_low	0.0129	0.0129	1
table_3	t3_ds_wcb_high	0.5718	0.5718	1
table_3	t3_ds_wcb_low	-0.3417	-0.3417	1
table_3	t3_no_ds_ci_lower	-1.6691	-1.6691	1
table_3	t3_no_ds_ci_upper	1.8359	1.8359	1
table_3	t3_no_ds_var_high	0.0059	0.0059	1
table_3	t3_no_ds_var_low	0.0062	0.0062	1
table_3	t3_no_ds_wcb_high	1.7097	1.7097	1
table_3	t3_no_ds_wcb_low	-1.5391	-1.5391	1
table_a1	ta1_all_ex_wave1_n	2156	2156	1
table_a1	ta1_all_ex_wave1_r2	0.001	0.001	1
table_a1	ta1_all_ex_wave2_n	2156	2156	1
table_a1	ta1_all_ex_wave2_r2	0.0004	0.0004	1
table_a1	ta1_all_pp_wave1_n	2156	2156	1
table_a1	ta1_all_pp_wave1_r2	0.024	0.024	1
table_a1	ta1_all_pp_wave2_n	2156	2156	1
table_a1	ta1_all_pp_wave2_r2	0.002	0.002	1
table_a1	ta1_constant_ex_wave1_coef	1.531	1.531	1
table_a1	ta1_constant_ex_wave1_se	0.024	0.024	1
table_a1	ta1_constant_ex_wave2_coef	1.617	1.617	1
table_a1	ta1_constant_ex_wave2_se	0.024	0.024	1
table_a1	ta1_constant_pp_wave1_coef	3.108	3.108	1
table_a1	ta1_constant_pp_wave1_se	0.046	0.046	1
table_a1	ta1_constant_pp_wave2_coef	3.542	3.542	1
table_a1	ta1_constant_pp_wave2_se	0.046	0.046	1
table_a1	ta1_placebo_ex_wave1_coef	0.019	0.019	1
table_a1	ta1_placebo_ex_wave1_se	0.035	0.035	1
table_a1	ta1_placebo_ex_wave2_coef	-0.011	-0.011	1
table_a1	ta1_placebo_ex_wave2_se	0.034	0.034	1
table_a1	ta1_placebo_pp_wave1_coef	0.352	0.352	1
table_a1	ta1_placebo_pp_wave1_se	0.067	0.067	1
table_a1	ta1_placebo_pp_wave2_coef	0.042	0.042	1
table_a1	ta1_placebo_pp_wave2_se	0.066	0.066	1
table_a1	ta1_polarized_ex_wave1_coef	-0.031	-0.031	1

Table 4: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity, or that no archive counterpart to that quantity was measured. *(continued)*

Location	Claim	Paper	Archive	Match
table_a1	ta1_polarized_ex_wave1_se	0.034	0.034	1
table_a1	ta1_polarized_ex_wave2_coef	-0.030	-0.030	1
table_a1	ta1_polarized_ex_wave2_se	0.034	0.034	1
table_a1	ta1_polarized_pp_wave1_coef	0.468	0.468	1
table_a1	ta1_polarized_pp_wave1_se	0.067	0.067	1
table_a1	ta1_polarized_pp_wave2_coef	0.126	0.126	1
table_a1	ta1_polarized_pp_wave2_se	0.066	0.066	1
table_a2	ta2_mod_vs_placebo_ex_ci_lower	-0.4778	-0.4778	1
table_a2	ta2_mod_vs_placebo_ex_ci_upper	0.5769	0.5769	1
table_a2	ta2_mod_vs_placebo_ex_var_high	0.0067	0.0067	1
table_a2	ta2_mod_vs_placebo_ex_var_low	0.0083	0.0083	1
table_a2	ta2_mod_vs_placebo_ex_wcb_high	0.4426	0.4426	1
table_a2	ta2_mod_vs_placebo_ex_wcb_low	-0.3284	-0.3284	1
table_a2	ta2_mod_vs_placebo_pp_ci_lower	-0.6119	-0.6119	1
table_a2	ta2_mod_vs_placebo_pp_ci_upper	0.5044	0.5044	1
table_a2	ta2_mod_vs_placebo_pp_var_high	0.0114	0.0114	1
table_a2	ta2_mod_vs_placebo_pp_var_low	0.0106	0.0106	1
table_a2	ta2_mod_vs_placebo_pp_wcb_high	0.3286	0.3286	1
table_a2	ta2_mod_vs_placebo_pp_wcb_low	-0.4424	-0.4424	1
table_a2	ta2_pol_vs_mod_ex_ci_lower	-0.6087	-0.6087	1
table_a2	ta2_pol_vs_mod_ex_ci_upper	0.6032	0.6032	1
table_a2	ta2_pol_vs_mod_ex_var_high	0.0095	0.0095	1
table_a2	ta2_pol_vs_mod_ex_var_low	0.0070	0.0070	1
table_a2	ta2_pol_vs_mod_ex_wcb_high	0.4426	0.4426	1
table_a2	ta2_pol_vs_mod_ex_wcb_low	-0.4710	-0.4710	1
table_a2	ta2_pol_vs_mod_pp_ci_lower	-0.5283	-0.5283	1
table_a2	ta2_pol_vs_mod_pp_ci_upper	0.7452	0.7452	1
table_a2	ta2_pol_vs_mod_pp_var_high	0.0111	0.0111	1
table_a2	ta2_pol_vs_mod_pp_var_low	0.0129	0.0129	1
table_a2	ta2_pol_vs_mod_pp_wcb_high	0.5718	0.5718	1
table_a2	ta2_pol_vs_mod_pp_wcb_low	-0.3417	-0.3417	1
table_a2	ta2_pol_vs_placebo_ex_ci_lower	-0.6004	-0.6004	1
table_a2	ta2_pol_vs_placebo_ex_ci_upper	0.6944	0.6944	1
table_a2	ta2_pol_vs_placebo_ex_var_high	0.0094	0.0094	1
table_a2	ta2_pol_vs_placebo_ex_var_low	0.0085	0.0085	1
table_a2	ta2_pol_vs_placebo_ex_wcb_high	0.5350	0.5350	1
table_a2	ta2_pol_vs_placebo_ex_wcb_low	-0.4491	-0.4491	1
table_a2	ta2_pol_vs_placebo_pp_ci_lower	-0.6171	-0.6171	1
table_a2	ta2_pol_vs_placebo_pp_ci_upper	0.7262	0.7262	1
table_a2	ta2_pol_vs_placebo_pp_var_high	0.0114	0.0114	1
table_a2	ta2_pol_vs_placebo_pp_var_low	0.0124	0.0124	1
table_a2	ta2_pol_vs_placebo_pp_wcb_high	0.5502	0.5502	1
table_a2	ta2_pol_vs_placebo_pp_wcb_low	-0.4339	-0.4339	1

Of the 171 recorded claims, 131 can be compared against the deposited scripts. 128 match the published value and 3 are unreachable, all because of the stale object name.

7 Maintained rewrite

The rewrite lives in `maintained/`: nine scripts covering the three main text tables, the two appendix tables, both figures, the numbers Sections 4.2 and 4.3 state in prose, and the claim-by-claim audit. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used, and the `attrition` functions that implement the method

(`estimator_ev`, `estimator_ds`, `sensitivity_ds`, and the internal `ds_manski_cis_2s` the simulation calls) are used unchanged, since they are the paper's contribution rather than incidental tooling.

Table 5: Scripts in the maintained rewrite.

Script	Produces
helpers.R	packages, data, shared row labels
table_1_attrition.R	Table 1: attrition by condition
table_2_outcome_means.R	Table 2: outcome means by condition and response type
table_3_bounds_main.R	Table 3: bounds with and without double sampling
table_a1_mipo_effects.R	Appendix Table 1: effects assuming MIPO
table_a2_pairwise_bounds.R	Appendix Table 2: bounds for all pairwise comparisons
figure_1_simulation.R	Figure 1 and Appendix Figure 1: the simulation study
figure_2_sensitivity.R	Figure 2: the sensitivity analysis
text_section_4_claims.R	the prose numbers in Sections 4.2 and 4.3
in_text_claims.R	one printed line per published claim, recomputed independently

Two structural choices are worth stating. `figure_1_simulation.R` writes both Figure 1 and Appendix Figure 1, which differ only in whether the larger sample size is shown, because they come from one simulation grid and splitting them would mean running it twice. `text_section_4_claims.R` reads the CSV output of the table and figure scripts rather than refitting anything, so a number quoted in the text and the same number printed in a table cannot come apart.

7.1 Deprecated patterns replaced

Table 6: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd(””)‘	‘here::here()‘
‘read.csv()‘	‘readr::read_csv()‘
‘lm()‘ + ‘commarobust::makerobustseslist()‘	‘estimatr::lm_robust(se_type = ”HC2”)‘
‘stargazer‘ + ‘out =‘	‘modelsummary(output = ...)‘
‘xtable‘	‘knitr::kable()‘ + ‘readr::write_lines()‘
‘multiplot()‘ (hand-rolled, grid-based)	‘gridExtra::arrangeGrob()‘
triple ‘for‘ loop building a matrix with ‘rbind‘	‘expand_grid()‘ + ‘pmap()‘ + ‘list_rbind()‘
‘reshape2::melt()‘	‘pivot_longer()‘
‘ifelse()‘	‘if_else()‘
‘data.frame()‘	‘tibble()‘
‘scale_color_grey(guide = FALSE)‘	‘scale_color_grey(guide = ”none”)‘
‘<ggplot> %+% data‘	‘<ggplot> + data‘
magrittr pipe	native pipe

`commarobust::commarobust()` is a wrapper around `lmtest::coeftest(fit, sandwich::vcovHC(fit, type = "HC2"))`, and `estimatr::lm_robust()` defaults to the same HC2 estimator, so the

substitution is exact: the standard errors agree to fifteen digits. The one visible difference is in the significance stars. `commarobust::makerobustpslist()` overwrites the intercept's p-value with 1 before handing the models to `stargazer`, which is why the published Appendix Table 1 shows no stars on the constant. The rewrite reports the intercept's actual p-value, so `table_a1_mipo_effects.tex` stars the constant where the published table does not. No estimate or standard error differs.

The simulation deserves a note of its own. The archive passes four objects named `c1a_t`, `c1r_t`, `c2a_t` and `c2r_t` into `ds_manski_cis_2s()` without ever defining them. This does not error, because they map onto formal arguments the function body never touches and R does not evaluate an argument it does not use. They are vestigial cost parameters from an earlier version of the function. The current function takes the arm-level moments as one list and has no such arguments; the rewrite builds that list directly.

8 Figures

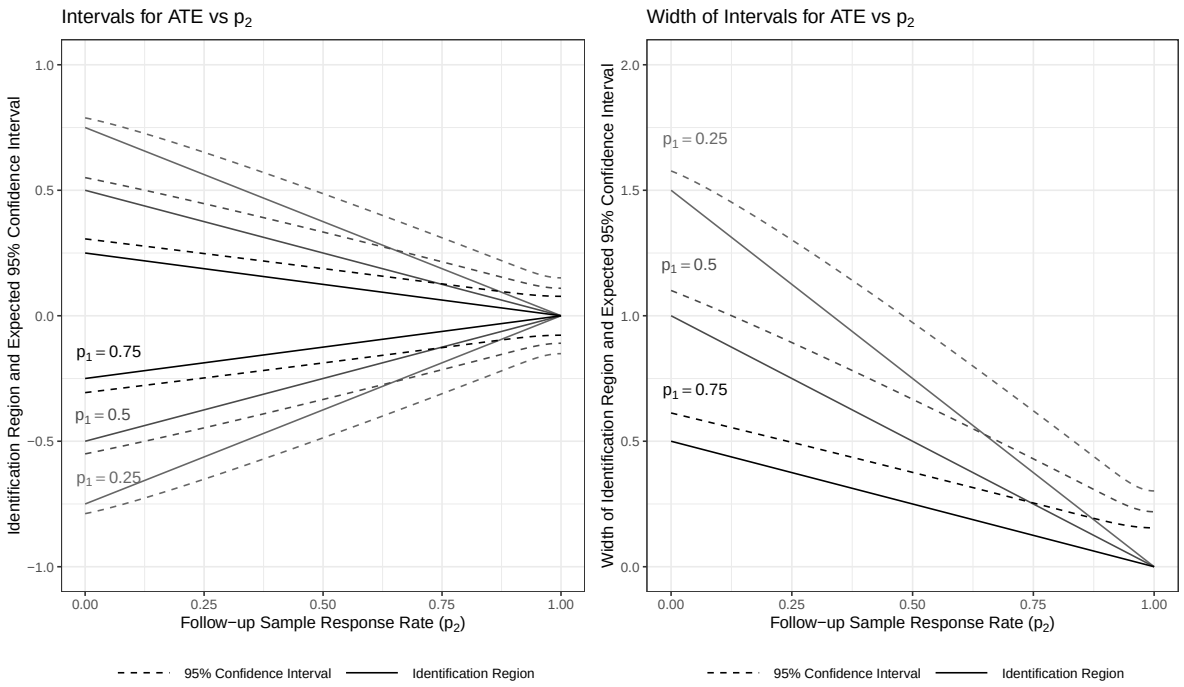


Figure 1: Figure 1 as reproduced by the maintained rewrite: the identification region and expected 95 percent confidence interval for the ATE against the follow-up response rate, at three initial response rates.

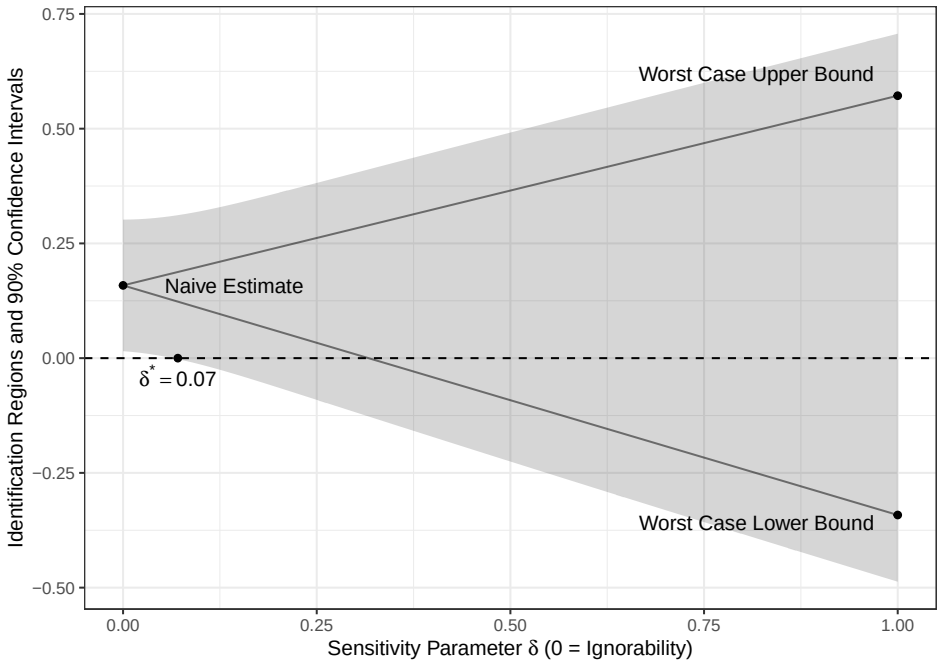


Figure 2: Figure 2 as reproduced by the maintained rewrite: the identification region and 90 percent confidence interval against the sensitivity parameter, with the delta star at which the interval first covers zero.

Neither figure carries printed estimates, so there is no panel label to check the way there is in some archives. Figure 2 is the exception that matters: it is annotated with $\delta^* = 0.07$, and that annotation is the only place the paper states the sensitivity result. The rewrite rebuilds both figures from its own data frames rather than saving the `ggplot` object `sensitivity_ds()` returns, so everything drawn here is produced by code in this repository. `figure_1_simulation.csv` carries the 6,000 point simulation grid behind Figure 1, which is what makes that figure diffable rather than merely inspectable.

9 Maintained rewrite verification

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number.

Location	Claim	Paper	Rewrite	Match	Holds
appendix_section_1_p2	appx_n_large	80000	80000	1	
appendix_section_1_p2	appx_n_small	800	800	1	
appendix_section_1_p2	appx_n2_large	5000	5000	1	
appendix_section_1_p2	appx_n2_small	50	50	1	
appendix_section_1_p2	appx_size_factor	100	100	1	
appendix_section_1_p2	appx_two_scenarios	2	2	1	

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number. *(continued)*

Location	Claim	Paper	Rewrite	Match	Holds
appendix_section_2_p3	appx_effect_diminished				1
appendix_section_2_p3	appx_extremity_negligible				1
appendix_section_2_p3	appx_wave1_effect	0.468	0.468375086098889	1	
appendix_section_2_p3	appx_wave1_effect_strong				1
appendix_section_2_p3	appx_wave2_effect	0.126	0.125886650690804	1	
appendix_section_3_p4	appx_pairwise_columns	6	6	1	
figure_1	sim_p2_zero_is_wcb				1
figure_1	sim_point_identified				1
figure_1	sim_region_smaller_in_p1				1
figure_1	sim_width_shrinks_to_zero				1
figure_2	fig2_delta_star	0.07	0.0707070707070707	1	
figure_a1	appx_ci_shrinks_region_does_not				1
section_3_p11	sim_n_followup	50	50	1	
section_3_p11	sim_n_total	800	800	1	
section_3_p11	sim_p1_max	0.75	0.75	1	
section_3_p11	sim_p1_min	0.25	0.25	1	
section_3_p11	sim_p2_max	1	1	1	
section_3_p11	sim_p2_min	0	0	1	
section_3_p11	sim_true_ate	0	0	1	
section_3_p12	sim_n_scenarios	3	3	1	
section_3_p12	sim_p1_scenario_high	0.75	0.75	1	
section_3_p12	sim_p1_scenario_low	0.25	0.25	1	
section_3_p12	sim_p1_scenario_mid	0.50	0.5	1	
section_4.1_p14	app_n_completed	72	72	1	
section_4.1_p14	app_n_invited	100	100	1	
section_4.1_p14	app_n_invited_per_arm	50	50	1	
section_4.1_p14	app_n_responded	1444	1444	1	
section_4.1_p14	app_n_subjects	1980	1980	1	
section_4.2_p14	app_outcome_max	6	6	1	
section_4.2_p14	app_outcome_min	0	-1.55889440680195e-15	1	
section_4.2_p14	txt_ci_width_ds_ps	1.23	1.22561085834396	1	
section_4.2_p14	txt_ci_width_no_ds	3.50	3.50496018440374	1	
section_4.2_p14	txt_ds_ps_ci_lower	-0.53	-0.529010551320638	1	
section_4.2_p14	txt_ds_ps_ci_upper	0.70	0.696600307023322	1	
section_4.2_p14	txt_ds_ps_wcb_high	0.53	0.525683688852529	1	
section_4.2_p14	txt_ds_ps_wcb_low	-0.34	-0.344389910863888	1	
section_4.2_p14	txt_ds_wcb_high	0.57	0.571815728388134	1	
section_4.2_p14	txt_ds_wcb_low	-0.34	-0.34174537662934	1	
section_4.2_p14	txt_naive_ate_coef	0.126	0.125886650690804	1	
section_4.2_p14	txt_naive_ate_pval	0.057	0.0571779950234041	1	
section_4.2_p14	txt_naive_ate_se	0.066	0.0661527954807814	1	
section_4.2_p14	txt_no_ds_ci_lower	-1.67	-1.66907903885775	1	
section_4.2_p14	txt_no_ds_ci_upper	1.84	1.83588114554598	1	
section_4.2_p14	txt_no_ds_wcb_high	1.71	1.70966762747749	1	
section_4.2_p14	txt_no_ds_wcb_low	-1.54	-1.53914496339566	1	
section_4.2_p14	txt_reduction_pct	0.65	0.650321032519101	1	
section_4.3_p15	app_naive_sig_10				1
section_4.3_p15	app_no_delta_star_05				1
section_4.3_p15	txt_delta_star	0.07	0.0707070707070707	1	
table_1	t1_moderate_did_not_respond	264	264	1	
table_1	t1_moderate_invited	50	50	1	
table_1	t1_moderate_n_initial	995	995	1	
table_1	t1_moderate_responded	731	731	1	
table_1	t1_moderate_responded_to_invitation	39	39	1	
table_1	t1_polarized_did_not_respond	272	272	1	
table_1	t1_polarized_invited	50	50	1	
table_1	t1_polarized_n_initial	985	985	1	
table_1	t1_polarized_responded	713	713	1	
table_1	t1_polarized_responded_to_invitation	33	33	1	
table_1	t1_total_did_not_respond	536	536	1	
table_1	t1_total_invited	100	100	1	
table_1	t1_total_n_initial	1980	1980	1	
table_1	t1_total_responded	1444	1444	1	
table_1	t1_total_responded_to_invitation	72	72	1	

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number. *(continued)*

Location	Claim	Paper	Rewrite	Match	Holds
table_2	app_samples_similar				1
table_2	t2_moderate_ds_mean	3.583	3.58333333333333	1	
table_2	t2_moderate_ds_n	39	39	1	
table_2	t2_moderate_ds_sd	1.367	1.36730301830269	1	
table_2	t2_moderate_ir_mean	3.542	3.54206566347469	1	
table_2	t2_moderate_ir_n	731	731	1	
table_2	t2_moderate_ir_sd	1.243	1.2426600835923	1	
table_2	t2_polarized_ds_mean	3.826	3.82575757575758	1	
table_2	t2_polarized_ds_n	33	33	1	
table_2	t2_polarized_ds_sd	1.313	1.31322130252061	1	
table_2	t2_polarized_ir_mean	3.668	3.6679523141655	1	
table_2	t2_polarized_ir_n	713	713	1	
table_2	t2_polarized_ir_sd	1.270	1.27045080160387	1	
table_3	app_ds_narrows				1
table_3	app_wcb_uninformative				1
table_3	t3_ds_ci_lower	-0.5283	-0.52830967410789	1	
table_3	t3_ds_ci_upper	0.7452	0.745174826433894	1	
table_3	t3_ds_ps_ci_lower	-0.5290	-0.529010551320638	1	
table_3	t3_ds_ps_ci_upper	0.6966	0.696600307023322	1	
table_3	t3_ds_ps_var_high	0.0108	0.0107972726598501	1	
table_3	t3_ds_ps_var_low	0.0126	0.0125981273119329	1	
table_3	t3_ds_ps_wcb_high	0.5257	0.525683688852529	1	
table_3	t3_ds_ps_wcb_low	-0.3444	-0.344389910863888	1	
table_3	t3_ds_var_high	0.0111	0.0111080739914738	1	
table_3	t3_ds_var_low	0.0129	0.0128647858310974	1	
table_3	t3_ds_wcb_high	0.5718	0.571815728388134	1	
table_3	t3_ds_wcb_low	-0.3417	-0.34174537662934	1	
table_3	t3_no_ds_ci_lower	-1.6691	-1.66907903885775	1	
table_3	t3_no_ds_ci_upper	1.8359	1.83588114554598	1	
table_3	t3_no_ds_var_high	0.0059	0.00588785669626298	1	
table_3	t3_no_ds_var_low	0.0062	0.0062401008346893	1	
table_3	t3_no_ds_wcb_high	1.7097	1.70966762747749	1	
table_3	t3_no_ds_wcb_low	-1.5391	-1.53914496339566	1	
table_a1	ta1_all_ex_wave1_n	2156	2156	1	
table_a1	ta1_all_ex_wave1_r2	0.001	0.000969193379921518	1	
table_a1	ta1_all_ex_wave2_n	2156	2156	1	
table_a1	ta1_all_ex_wave2_r2	0.0004	0.000359806687159292	1	
table_a1	ta1_all_pp_wave1_n	2156	2156	1	
table_a1	ta1_all_pp_wave1_r2	0.024	0.0239507001082565	1	
table_a1	ta1_all_pp_wave2_n	2156	2156	1	
table_a1	ta1_all_pp_wave2_r2	0.002	0.00174424183477495	1	
table_a1	ta1_constant_ex_wave1_coef	1.531	1.53146374829	1	
table_a1	ta1_constant_ex_wave1_se	0.024	0.0243596218569332	1	
table_a1	ta1_constant_ex_wave2_coef	1.617	1.61662106703145	1	
table_a1	ta1_constant_ex_wave2_se	0.024	0.024150796568631	1	
table_a1	ta1_constant_pp_wave1_coef	3.108	3.10841313269491	1	
table_a1	ta1_constant_pp_wave1_se	0.046	0.046462289784369	1	
table_a1	ta1_constant_pp_wave2_coef	3.542	3.54206566347466	1	
table_a1	ta1_constant_pp_wave2_se	0.046	0.0459614434759558	1	
table_a1	ta1_placebo_ex_wave1_coef	0.019	0.0190980494627954	1	
table_a1	ta1_placebo_ex_wave1_se	0.035	0.035282865017696	1	
table_a1	ta1_placebo_ex_wave2_coef	-0.011	-0.010581741188767	1	
table_a1	ta1_placebo_ex_wave2_se	0.034	0.0343376300527564	1	
table_a1	ta1_placebo_pp_wave1_coef	0.352	0.352378065806935	1	
table_a1	ta1_placebo_pp_wave1_se	0.067	0.067434267815023	1	
table_a1	ta1_placebo_pp_wave2_coef	0.042	0.041501752255645	1	
table_a1	ta1_placebo_pp_wave2_se	0.066	0.0656684123731185	1	
table_a1	ta1_polarized_ex_wave1_coef	-0.031	-0.0311131171539694	1	
table_a1	ta1_polarized_ex_wave1_se	0.034	0.034340859741032	1	
table_a1	ta1_polarized_ex_wave2_coef	-0.030	-0.0296645452923338	1	
table_a1	ta1_polarized_ex_wave2_se	0.034	0.03402111975226	1	
table_a1	ta1_polarized_pp_wave1_coef	0.468	0.468375086098889	1	
table_a1	ta1_polarized_pp_wave1_se	0.067	0.0665244501080027	1	
table_a1	ta1_polarized_pp_wave2_coef	0.126	0.125886650690804	1	

Table 7: Maintained rewrite verification: published value against rewrite output. A row with a Holds verdict and no value is a claim about shape rather than about a number. *(continued)*

Location	Claim	Paper	Rewrite	Match	Holds
table_a1	ta1_polarized_pp_wave2_se	0.066	0.0661527954807814	1	
table_a2	ta2_mod_vs_placebo_ex_ci_lower	-0.4778	-0.477842687179526	1	
table_a2	ta2_mod_vs_placebo_ex_ci_upper	0.5769	0.57692983337372	1	
table_a2	ta2_mod_vs_placebo_ex_var_high	0.0067	0.00666486134438053	1	
table_a2	ta2_mod_vs_placebo_ex_var_low	0.0083	0.0082562481709875	1	
table_a2	ta2_mod_vs_placebo_ex_wcb_high	0.4426	0.442646282695529	1	
table_a2	ta2_mod_vs_placebo_ex_wcb_low	-0.3284	-0.328384873083365	1	
table_a2	ta2_mod_vs_placebo_pp_ci_lower	-0.6119	-0.611862038531696	1	
table_a2	ta2_mod_vs_placebo_pp_ci_upper	0.5044	0.504351636768822	1	
table_a2	ta2_mod_vs_placebo_pp_var_high	0.0114	0.0114143454384199	1	
table_a2	ta2_mod_vs_placebo_pp_var_low	0.0106	0.0106127353002281	1	
table_a2	ta2_mod_vs_placebo_pp_wcb_high	0.3286	0.328618863548512	1	
table_a2	ta2_mod_vs_placebo_pp_wcb_low	-0.4424	-0.442412292230382	1	
table_a2	ta2_pol_vs_mod_ex_ci_lower	-0.6087	-0.608731543014259	1	
table_a2	ta2_pol_vs_mod_ex_ci_upper	0.6032	0.603194110577457	1	
table_a2	ta2_pol_vs_mod_ex_var_high	0.0095	0.00953316194586098	1	
table_a2	ta2_pol_vs_mod_ex_var_low	0.0070	0.00701487544541756	1	
table_a2	ta2_pol_vs_mod_ex_wcb_high	0.4426	0.442594036170701	1	
table_a2	ta2_pol_vs_mod_ex_wcb_low	-0.4710	-0.470967068846773	1	
table_a2	ta2_pol_vs_mod_pp_ci_lower	-0.5283	-0.52830967410789	1	
table_a2	ta2_pol_vs_mod_pp_ci_upper	0.7452	0.745174826433894	1	
table_a2	ta2_pol_vs_mod_pp_var_high	0.0111	0.0111080739914738	1	
table_a2	ta2_pol_vs_mod_pp_var_low	0.0129	0.0128647858310974	1	
table_a2	ta2_pol_vs_mod_pp_wcb_high	0.5718	0.571815728388134	1	
table_a2	ta2_pol_vs_mod_pp_wcb_low	-0.3417	-0.34174537662934	1	
table_a2	ta2_pol_vs_placebo_ex_ci_lower	-0.6004	-0.600425131607832	1	
table_a2	ta2_pol_vs_placebo_ex_ci_upper	0.6944	0.694385360492085	1	
table_a2	ta2_pol_vs_placebo_ex_var_high	0.0094	0.00938841779116446	1	
table_a2	ta2_pol_vs_placebo_ex_var_low	0.0085	0.00846151811732802	1	
table_a2	ta2_pol_vs_placebo_ex_wcb_high	0.5350	0.535009163087335	1	
table_a2	ta2_pol_vs_placebo_ex_wcb_low	-0.4491	-0.449120786151244	1	
table_a2	ta2_pol_vs_placebo_pp_ci_lower	-0.6171	-0.617118472710443	1	
table_a2	ta2_pol_vs_placebo_pp_ci_upper	0.7262	0.726201234202598	1	
table_a2	ta2_pol_vs_placebo_pp_var_high	0.0114	0.0114487996375541	1	
table_a2	ta2_pol_vs_placebo_pp_var_low	0.0124	0.012403901338986	1	
table_a2	ta2_pol_vs_placebo_pp_wcb_high	0.5502	0.550203436157751	1	
table_a2	ta2_pol_vs_placebo_pp_wcb_low	-0.4339	-0.433926513080827	1	

Every comparable row carries `match_rewrite = 1`: **158** values produced by the maintained rewrite, all matching the published paper to the precision the page prints, and **13** descriptive claims that hold. Every value here is read out of a file in `maintained/output/` by `build_ground_truth.R` rather than transcribed, so a value in this table that disagreed with the pipeline would be a failure of the pipeline and not of the transcription.

10 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23

Date run	2026-09-27

Table 9: Package versions used for the run behind this report.

Package	Version
attrition	1.0.0
estimatr	2.0.1
modelsummary	2.6.0
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
here	1.0.2
gridExtra	2.3
knitr	1.51
kableExtra	1.4.0

Table 10: The deposited files, as pinned in the manifest. All seven served checksums match the checksums Dataverse publishes, and all seven match the local copy this work was done against.

File	Bytes	MD5 (served)	Published MD5 agrees
README.txt	2090	ea327d0299d36ebd0f5bc1b4a01e5c6d	yes
attrition_0.0.0.9000.tar.gz	34511	7fc29b91569bff874735191147a65fc8	yes
commarobust_0.1.0.tar.gz	8085	feef406b22d3259c23d16f91ffbaadc9	yes
experimental_analysis_appendix.R	3756	5fc6e7d1fe25133820d96648ddb7ebfb	yes
experimental_analysis_main_text.R	3668	19a3cf55e8685f65d23419cd013fae0e	yes
levendusky_mturk_clean.csv	178043	6290d4802c5331dfa550fc954ccf07d0	yes
simulation_analysis.R	6358	ebdc8595bddfdef33ee8dde128366039	yes